

Final Exam

CS 5350 Parallel and Distributed Algorithms

Due Date: 05/12 11:59PM (updated)

PROJECT DESCRIPTION

1. Design and implement parallel algorithms for a Sudoku solver.

- **Problem statement:** Give an incomplete sudoku board, the parallel algorithms should provide a solution within finite time.
- You will develop the following algorithms to solve a given sudoku board
 - a) A brute force algorithm [A]: This algorithm will look at all possible combinations for the Sudoku board, and identify one possible solution.
 - b) Efficient algorithm [B]: You will conduct a study to see if there is any existing algorithms or heuristics that have been developed to solve Sudoku boards more efficiently than the brute force approach. For example, see the [Crook's algorithm](#). You will select one such algorithm, and design and implement the parallel version.
- You may use either MPI or OpenMP or a combination of both to implement the algorithms
- You will conduct performance comparison of the serial and parallel versions of algorithms A and B using appropriate using both theoretical and experimental analysis. You will utilize appropriate performance metrics.

2. Deliverables:

All submissions, unless otherwise specified, will be done through Blackboard using the link on canvas

Final Submission and presentation

Deadline for submission: 05/15 11:59 PM

- A PowerPoint file that provides a description of your design, analysis. You may use flowcharts, UML etc. to provide details of your implementation. You will also provide the tasks performed by each team member.
- The source code files.
- The compiled files
- Each member of the team will submit the slides (same copy) on blackboard using the link for “final submission” on canvas.

Each team will schedule a presentation with the instructor (the week of May 11th). Your presentation will last 20 minutes. The instructor will email you to set up the presentations.

Anyone who misses the final presentation will not receive a grade.